Problem

Title

Binary to Decimal Conversion (Manual Logic Implementation)

Topics

Number System Conversion, Positional Value Logic, Loops, Conditional Statements, Mathematical Computation

Difficulty Level

Medium

Problem Detailed Statement (Scenario Based)

In a computer fundamentals session, students are now given the reverse challenge of number conversion.

The trainer provides a **binary number (base 2)** and asks students to convert it into its **decimal equivalent (base 10)** without using built-in conversion functions.

Binary numbers consist only of digits **0 and 1**.

To convert binary to decimal manually, students must:

1. Read the binary number.
2. Starting from the rightmost digit (Least Significant Bit), assign powers of 2.
3. Multiply each digit by its corresponding power of 2.
4. Sum all the results.

For example:

If binary number = 1010

Positions (from right to left):

$0 \times 2^0 = 0$
$1 \times 2^1 = 2$
$0 \times 2^2 = 0$
$1 \times 2^3 = 8$

Total = 8 + 2 = **10**

Another example:

Binary = 1101

$1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$
$= 8 + 4 + 0 + 1$
$= 13$

Students must:

- Use loop logic to extract digits.
- Avoid string-based direct conversion.
- Use mathematical power calculation (without built-in power function ideally).
- Validate that input contains only 0 and 1.
- Handle edge case where input is 0.

This problem strengthens:

- Positional number system understanding
- Loop-based computation
- Mathematical power logic
- Conditional validation
- Digit extraction reasoning

Input Format

A single integer representing a binary number:

`N`

Constraints:

`0 ≤ N ≤ 10^18`

Note: The number contains only digits 0 and 1.

Output Format

- Print a single integer representing the decimal equivalent.

Examples

Example 1

Input:

```
1010
```

Output:

```
10
```

Example 2

Input:

```
1101
```

Output:

```
13
```

Example 3

Input:

```
0
```

Output:

```
0
```

Example 4

Input:

```
111111
```

Output:

```
63
```